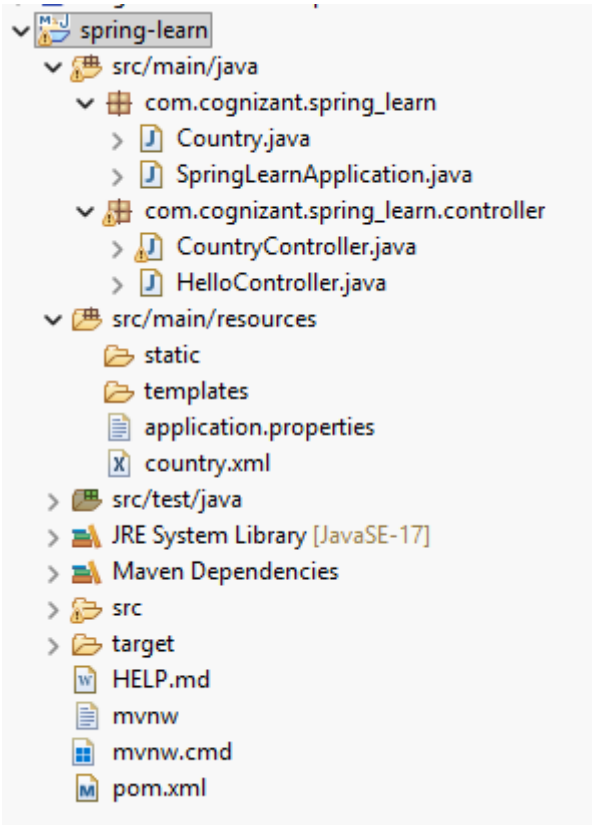


REST - Country Web Service



pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.3</version>
    <relativePath/>
  </parent>

  <groupId>com.cognizant</groupId>
  <artifactId>spring-learn</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name>spring-learn</name>

  <properties>
    <java.version>17</java.version>
  </properties>

  <dependencies>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
```

```

        <artifactId>spring-boot-devtools</artifactId>
        <scope>runtime</scope>
        <optional>true</optional>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>

    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-context</artifactId>
    </dependency>

</dependencies>

<build>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>
    </plugins>
</build>
</project>

```

application.properties:

```

spring.application.name=spring-learn
server.port=8083
logging.level.root=INFO
logging.level.com.cognizant.spring-learn=DEBUG

```

Country.java:

```

package com.cognizant.spring_learn;

public class Country {

    private String code;
    private String name;

    public Country() {
    }

    public String getCode() {
        return code;
    }

    public void setCode(String code) {
        this.code = code;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
}

```

Country.xml:

?? xml	version="1.0" encoding="UTF-8"
▼ beans	(description?, (import alias bean namespace:uri="##other"), beans*)
xmlns	http://www.springframework.org/schema/beans
xmlns:xsi	http://www.w3.org/2001/XMLSchema-instance
xsi:schemaLocation	http://www.springframework.org/schema/beans https://www.springframework.org/schema/beans/spring-beans.xsd
▼ bean	((description?, (meta constructor-arg property qualifier lookup-method replaced-method namespace:uri="##other")))
id	country
class	com.cognizant.spring_learn.Country
▼ property	(description?, (meta bean ref idref value null array list set map props namespace:uri="##other"))?
name	code
value	IN
▼ property	(description?, (meta bean ref idref value null array list set map props namespace:uri="##other"))?
name	name
value	India

CountryController.java:

```
package com.cognizant.spring_learn.controller;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.spring_learn.Country;

@RestController
public class CountryController {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(CountryController.class);

    @RequestMapping("/country")
    public Country getCountryIndia() {

        LOGGER.info("START");

        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");

        Country country =
            context.getBean("country", Country.class);

        LOGGER.info("END");

        return country;
    }
}
```

SpringLearnApplication.java:

```
package com.cognizant.spring_learn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

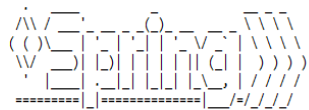
@SpringBootApplication

public class SpringLearnApplication {
```

```
public static void main(String[] args) {  
  
    SpringApplication.run(SpringLearnApplication.class, args);  
  
}  
}
```

OUTPUT:

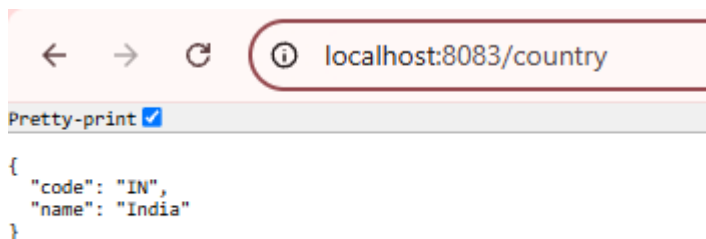
1) Console:



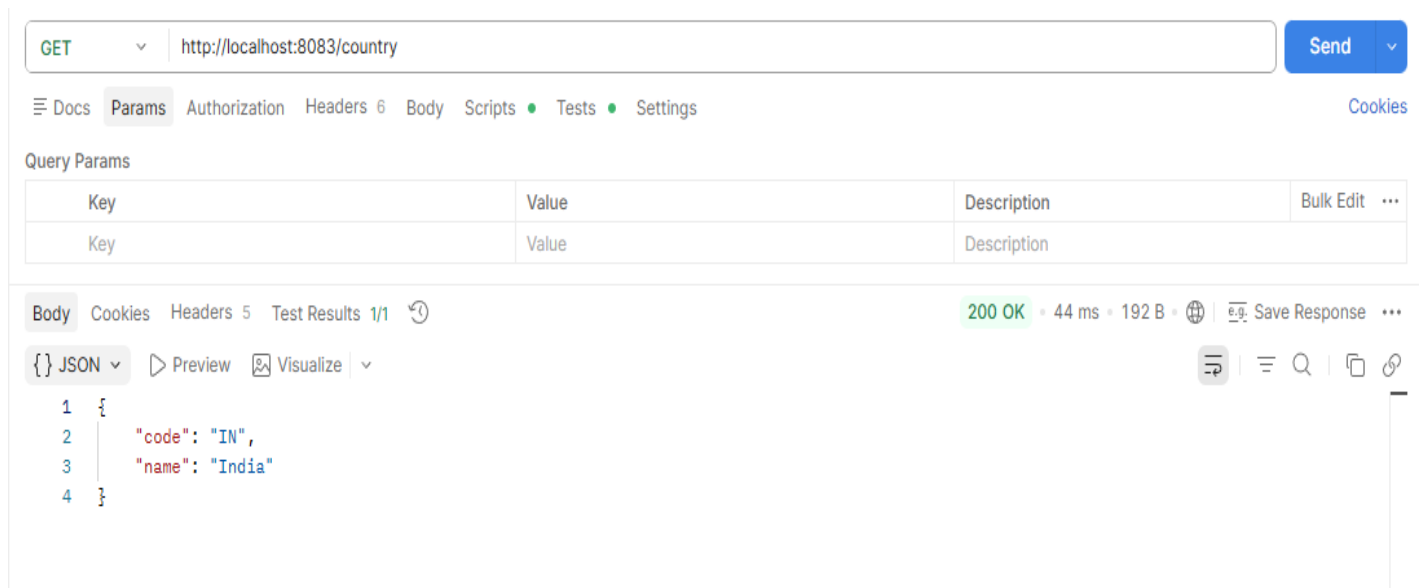
:: Spring Boot :: (v3.5.3)

```
2026-07-05T17:20:45.385+05:30 INFO 17196 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.9 with PID 17196 (C:\Users\A  
2026-07-05T17:20:45.388+05:30 INFO 17196 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"  
2026-07-05T17:20:45.429+05:30 INFO 17196 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to '  
2026-07-05T17:20:45.429+05:30 INFO 17196 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web'  
2026-07-05T17:20:46.200+05:30 INFO 17196 --- [spring-learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)  
2026-07-05T17:20:46.213+05:30 INFO 17196 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]  
2026-07-05T17:20:46.214+05:30 INFO 17196 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.42]  
2026-07-05T17:20:46.248+05:30 INFO 17196 --- [spring-learn] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext  
2026-07-05T17:20:46.249+05:30 INFO 17196 --- [spring-learn] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 819 ms  
2026-07-05T17:20:46.563+05:30 INFO 17196 --- [spring-learn] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729  
2026-07-05T17:20:46.598+05:30 INFO 17196 --- [spring-learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'  
2026-07-05T17:20:46.607+05:30 INFO 17196 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 1.566 seconds (process running for 1.886)  
2026-07-05T17:20:58.889+05:30 INFO 17196 --- [spring-learn] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'  
2026-07-05T17:20:58.889+05:30 INFO 17196 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'  
2026-07-05T17:20:58.891+05:30 INFO 17196 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms  
2026-07-05T17:20:58.922+05:30 INFO 17196 --- [spring-learn] [nio-8083-exec-1] c.c.s.controller.HelloController : START  
2026-07-05T17:20:58.922+05:30 INFO 17196 --- [spring-learn] [nio-8083-exec-1] c.c.s.controller.HelloController : END
```

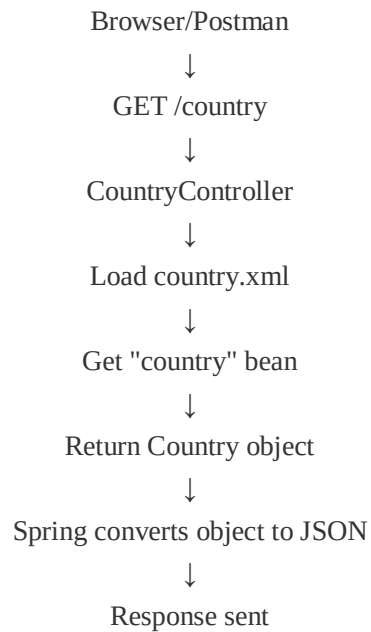
2) Browser:



3) Postman:



What happens in the controller method?



How the bean is converted into JSON response?

When the method returns:

```
return country;
```

Spring Boot uses the Jackson library (included through spring-boot-starter-web).

Internally:

```
Country country = new Country();
country.setCode("IN");
country.setName("India");
```

becomes:

```
{
  "code": "IN",
  "name": "India"
}
```

Spring automatically calls the getters:

```
getCode()
getName()
```

and creates JSON.

In network tab of developer tools , the HTTP header details received:

Name	X	Headers	Preview	Response	Initiator	Timing	Cookies
country							
General							
Request URL		http://localhost:8083/country					
Request Method		GET					
Status Code		200 OK					
Remote Address		[::1]:8083					
Referrer Policy		strict-origin-when-cross-origin					
Response headers Raw							
HTTP/1.1 200							
Content-Type: application/json							
Transfer-Encoding: chunked							
Date: Sun, 05 Jul 2026 12:05:55 GMT							
Keep-Alive: timeout=60							
Connection: keep-alive							
Request Headers Raw							
GET /country HTTP/1.1							
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exch=0.7							
Accept-Encoding: gzip, deflate, br, zstd							
Accept-Language: en-US,en;q=0.9							
Cache-Control: max-age=0							
Connection: keep-alive							
Cookie: ajs_anonymous_id=6a6dfe8b-23fc-4a8e-826f-f3a40ce992c1							
Host: localhost:8083							
Sec-Fetch-Dest: document							
Sec-Fetch-Mode: navigate							
Sec-Fetch-Site: none							
Sec-Fetch-User: ?1							
Upgrade-Insecure-Requests: 1							
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/149.0.0.0 Safari/537.36							
sec-ch-ua: "Google Chrome";v="149", "Chromium";v="149", "Not)A;Brand";v="24"							
sec-ch-ua-mobile: ?0							
sec-ch-ua-platform: "Windows"							

In postman , the HTTP header details received in "Headers" tab:

Body Cookies Headers 5 Test Results 1/1 200 OK • 71 ms • 192 B • Save Response ...

Key	Value
Content-Type	application/json
Transfer-Encoding	chunked
Date	Sun, 05 Jul 2026 12:04:43 GMT
Keep-Alive	timeout=60
Connection	keep-alive